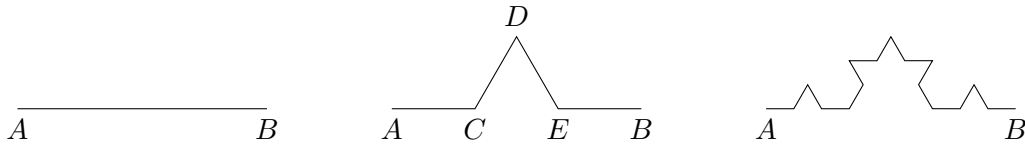


The Koch Snowflake

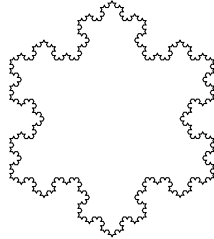
1 Definition

The *Koch curve* of depth n between two points A and B is defined recursively. For $n = 0$ it is simply the segment $[A, B]$. For $n > 0$, the segment is divided into three equal parts $[A, C]$, $[C, E]$, $[E, B]$, and the middle part is replaced by the two upper sides of the equilateral triangle CED erected on it. The curve of depth n then consists of four Koch curves of depth $n - 1$, along $[A, C]$, $[C, D]$, $[D, E]$ and $[E, B]$.



Depths 0, 1 and 2 of the Koch curve.

The *Koch snowflake* is obtained by tracing three Koch curves along the three sides of an equilateral triangle, bumps pointing outward:



2 The geometry of one step

With points as vectors, the construction of one step is three formulas:

$$C = A + \frac{1}{3}(B - A), \quad E = A + \frac{2}{3}(B - A), \quad D = C + R_{60^\circ}(E - C), \quad (1)$$

where R_{60° is the rotation by 60° ,

$$R_\theta \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \cos \theta - y \sin \theta \\ x \sin \theta + y \cos \theta \end{pmatrix}. \quad (2)$$

The sign of the angle decides on which side of $[A, B]$ the bump grows; flipping it everywhere turns the snowflake into the *anti-snowflake*, whose bumps point inward.

3 Properties

Start from an equilateral triangle of side s and area $A_0 = \frac{\sqrt{3}}{4}s^2$.

- **Exponentially many segments.** Each step replaces one segment by four of one third the length: depth n has 4^n segments per side, each of length $s/3^n$.
- **Infinite perimeter.** The boundary length is $3s \left(\frac{4}{3}\right)^n \rightarrow \infty$: the limit curve has infinite length, yet fits in a bounded region.
- **Finite area.** Each step adds ever-smaller triangles; the total converges to $\frac{8}{5}A_0$. An infinite fence around a finite lawn.
- **Fractal dimension.** The limit curve is self-similar with ratio 3 producing 4 copies, so its Hausdorff dimension is $\log 4 / \log 3 \approx 1.262$ — strictly between a line and a plane.
- **Continuous, nowhere differentiable.** The limit curve has a corner in every neighborhood of every point; it was one of the early “pathological” curves (von Koch, 1904).

4 From mathematics to pixels

The program computes the depth- n curve recursively: a depth-0 call contributes its endpoint, a deeper call delegates to its four children. The recursion is a *tree* of branching factor four — unlike a loop-shaped (tail) recursion, its work grows as 4^n , which is why the drawing depth is capped: depth 6 already traces $3 \cdot 4^6 = 12,288$ segments, well below a pixel each at screen size.